

Below is a complete, self-contained project brief that includes every decision, rationale, and component we've discussed. It is written so that another AI can generate the entire project code, notebook, and report without missing any nuance.

PROJECT CONTEXT: RL-ESG Portfolio Optimization with Sentiment Integration

1. OVERALL OBJECTIVE

Extend **Feng et al. (2024)** – “Sustainable Portfolio Construction via Machine Learning: ESG, SDG and Sentiment” – by replacing the paper’s static monthly ML allocation with a **dynamic reinforcement learning (RL) agent** that trades daily using NLP-derived sentiment scores. The system must:

- Use **only real, public data** (no synthetic data)
- Construct **daily ESG, SDG, and Overall sentiment indicators** from financial news
- Pre-select stocks using a two-stage filter: sentiment → Fama-French 3-factor alpha
- Train an RL agent (PPO) to output portfolio weights daily, optimizing a **coherent downside risk measure (CVaR)** instead of symmetric variance
- Outperform the paper’s benchmarks (equal-weight, monthly mean-variance) after transaction costs
- Be academically rigorous, mathematically sound, and fully reproducible

2. ORIGINAL PAPER SUMMARY (Feng et al. 2024)

What they did:

- Collected 17M+ news articles for all SPX500 & STOXX600 firms (Feb 2015 – Feb 2022)
- Used **lexicon-based NLP** to compute daily sentiment tone (range -100 to +100)
- Tagged articles as ESG-related or SDG-related using ElasticSearch topic queries
- Created three daily sentiment scores per stock: **Overall, ESG, SDG**
- Each month, selected top-M stocks by 6-month average sentiment (M=10 or 30)
- Used machine learning models (RF, NN, GA, LASSO, etc.) to derive optimal weights via Mean-Variance optimization
- Benchmarked against equal-weighted portfolios; showed nonlinear models (RF, NN, GA) perform best

Limitations we address:

- Static monthly re-optimization ignores intra-month news dynamics
- Variance is a symmetric risk measure (penalizes upside)
- No factor-adjusted screening (sentiment may just capture market beta)
- No transaction-cost-aware dynamic trading

3. OUR EXTENSION & CONTRIBUTIONS

1. **Daily RL Agent** – learns a policy that adjusts weights every trading day based on fresh sentiment and market state

2. **Asymmetric Coherent Risk** – replace variance with CVaR (Conditional Value-at-Risk) in the reward function; also reward upside explicitly
3. **Factor-Adjusted Stock Selection** – after sentiment screening, select stocks with the highest **Fama-French 3-factor alpha** (not just Jensen's Alpha) to remove size/value effects
4. **Improved Sentiment Pipeline** – combine BM25 relevance scoring with FinBERT sentiment, outperforming simple lexicon/word-count methods
5. **Fully Reproducible Real Data** – use FNSPID (15.7M news, 1999-2023) + yfinance prices

4. DATA SOURCES & RATIONALE

Stock Prices, Index, Risk-Free Rate:

- Source: `yfinance` (free, no API key)
- Tickers: S&P 500 constituents (from Wikipedia, cleaned)
- Period: 2018-01-01 to 2025-12-31
- Daily adjusted close prices → log returns
- Also download S&P 500 index (^GSPC) and 1-month T-bill (^IRX) for factor models
- Filter out stocks with >30% missing data and zero-variance (dead stocks)
- **Result:** 488 stocks with clean daily log returns (saved)

Fama-French 3-Factor Data:

- Source: Ken French Data Library (via `pandas\_datareader`)
- Daily frequency, aligned with stock returns
- Used to compute rolling 252-day 3-factor alphas for each stock (already computed and saved)

News Data (FNSPID):

- Source: **Financial News and Stock Price Integration Dataset** (Hugging Face: `Zihan1004/FNSPID`)
- File: `Stock\_news/nasdaq\_exteral\_data.csv` (~23 GB) – currently downloading
- Covers 4,775 S&P 500 firms, 1999–2023, from 4 major financial news websites
- Columns: `ticker`, `date`, `title` (headline), `source`
- We filter to our 488 tickers and 2018-2020 window (overlap with price data) to keep memory manageable
- **Why FNSPID over Benzinga:** FNSPID is multi-source, peer-reviewed, and spans 1999–2023, making it the closest public equivalent to the paper's proprietary corpus. Benzinga is single-source and ends mid-2020.

5. SENTIMENT & ESG/SDG SCORING PIPELINE (Friend's Advanced Approach)

We adopt a **two-layer scoring system** that improves upon the paper's simple lexicon:

Layer 1 – Topic Relevance via BM25:

- Build a small TF-IDF/BM25 index of ESG and SDG keyword phrases (derived from standard sustainability taxonomies and the UN SDG list)
- For each news headline (title) for a given stock:
 - Compute BM25 score against the ESG keyword set → `ESG\_BM25`
 - Compute BM25 score against the SDG keyword set → `SDG\_BM25`

- Apply threshold T: if `ESG\_BM25 > T`, set binary indicator `I\_ESG = 1`, else `0`; similarly for SDG
- This is more robust than simple substring matching because BM25 accounts for term frequency and document length normalization

****Layer 2 – Sentiment via FinBERT:****

- Use `ProsusAI/finbert` (pre-trained on financial text) to classify each headline as positive/negative/neutral
- Compute tone = `positive\_proba - negative\_proba` (range -1 to 1)

****Combined Daily Scores:****

- For each article about stock `i` on day `t`:
 - `ESG\_score\_article = I\_ESG \* tone`
 - `SDG\_score\_article = I\_SDG \* tone`
 - `Overall\_score\_article = tone` (regardless of topic)
- For each stock each day, average all articles:
 - `ESG\_score(i,t) = mean(ESG\_score\_article)`
 - `SDG\_score(i,t) = mean(SDG\_score\_article)`
 - `Overall\_score(i,t) = mean(Overall\_score\_article)`
- If no articles, score = 0 (as in paper)

****Implementation Notes:****

- FinBERT inference on GPU (RTX 4070) with batch size 64-128
- BM25 can be computed per article on-the-fly using a pre-tokenized ESG/SDG term list
- Because FNSPID gives pre-tagged tickers, we don't need to retrieve news per stock; we just filter the CSV. The BM25 step is only for ESG/SDG classification, not retrieval.

6. FACTOR-ADJUSTED STOCK SELECTION

After daily sentiment scores are built:

1. ****Rolling 6-month average sentiment**** (ESG + SDG + Overall combined) → rank stocks, pick top 30
2. From these 30, select ****top 10 by Fama-French 3-factor alpha**** (rolling 252-day, annualized)
 - Regression: $r_i - r_f = \alpha + \beta_1 (\text{Mkt-RF}) + \beta_2 \text{SMB} + \beta_3 \text{HML} + \epsilon$
 - Alpha = intercept × 252
 - This removes size and value premia from the selection, isolating true ESG-driven alpha

7. RL PORTFOLIO OPTIMIZATION FRAMEWORK

****Environment (gym.Env):****

- ****State**** (daily):
 - Past 20 daily returns for the 10 selected stocks (10×20 = 200)
 - Current day's ESG, SDG, Overall sentiment scores for the 10 stocks (10×3 = 30)
 - Previous portfolio weights (10)
 - Total state dim = 240

- **Action**: New portfolio weights (10-dim, long-only, sum=1) from a Dirichlet policy distribution
- **Reward**:

$$R_t = w_t^{\top} r_{t+1} - \lambda_1 \text{CVaR}_{95\%}(w_t^{\top} r_{\text{rolling}}) + \lambda_2 \max(0, w_t^{\top} r_{t+1} - \mu_{\text{rolling}}) - \tau \|w_t - w_{t-1}\|_1$$
 - λ_1 = risk aversion, τ = transaction cost coefficient, λ_2 = upside reward coefficient
 - CVaR computed from trailing 60-day portfolio returns
- **Episode**: one month of trading; at month end, stocks are re-selected via the two-stage filter
- **Constraints**: no short selling, weights sum to 1, max weight per stock ≤ 0.3 (optional)

RL Algorithm:

- PPO (Stable-Baselines3) with continuous action space
- Policy network and value network: MLPs with 2-3 hidden layers
- Training on 2018-2020 data (daily steps), testing out-of-sample on 2021-2022 (or 2025)
- Hyperparameter tuning via grid search on validation set (Sharpe ratio)

Benchmarks:

1. Equal-weight (1/10)
2. Monthly Mean-CVaR optimization (rolling 6-month window)
3. Monthly Random Forest allocation (replicate paper's best ML)

Evaluation Metrics:

- Out-of-sample annualized return, volatility, Sharpe ratio, Sortino ratio, max drawdown
- Average monthly turnover
- Bootstrap test for Sharpe difference significance

8. WHAT WE HAVE COMPLETED (STEP 0)

- S&P 500 tickers fetched (488 valid)
- Daily log returns, SPX excess returns, risk-free rate saved (`data/processed/log\_returns.csv`)
- Fama-French 3-factor daily data downloaded (`data/raw/ff3\_factors.csv`)
- Rolling market beta & Jensen's Alpha computed (`beta\_market.csv`, `alpha\_market.csv`)
- Rolling 3-factor alpha computed (`alpha\_3factor.csv`)
- Date range aligned (after dropping 252-day warm-up): approx 2019-2025, 488 stocks

9. CURRENT STATE

- **News data (FNSPID)** is downloading: `nasdaq\_exteral\_data.csv` (~23 GB) via direct HTTP. Expected time: 30-60 minutes.
- **Next immediate step** (when download finishes): filter the CSV to our 488 tickers and 2018-2020 window, keeping columns `ticker`, `date`, `title`.
- Then proceed with the full sentiment pipeline (BM25 + FinBERT).

10. DECISIONS & RATIONALES

- **FNSPID over Benzinga**: multi-source, academically validated, larger span, matches paper's spirit
- **3-factor Alpha over Jensen's Alpha**: removes size/value effects, a modern factor-neutral metric
- **CVaR over Variance**: coherent risk measure, penalizes only downside, satisfies sub-additivity etc.
- **BM25 instead of simple keyword match**: more sophisticated NLP, captures term importance, defensible
- **FinBERT instead of lexicon**: context-aware sentiment, state-of-the-art for financial text
- **RL (PPO) over static ML**: learns dynamic policy, optimizes for reward that includes turnover and risk directly
- **No Streamlit for now**: focus on core model; live dashboard can be added later as future work
- **Real data only**: no synthetic generation, all datasets publicly verifiable

11. PROJECT STRUCTURE (as built)

```

...
esg_rl_portfolio/
├── data/
│   ├── raw/          # raw downloads (stock_prices.csv, ff3_factors.csv, fnspid_news.csv)
│   └── processed/     # cleaned data (log_returns.csv, alpha_3factor.csv, stock_list.csv,
daily_sentiment_scores.csv)
├── notebooks/
│   ├── 00_data_setup.ipynb  # Step 0 completed
│   ├── 01_sentiment.ipynb   # Step 1 (to be created)
│   └── 02_rl_agent.ipynb    # Step 2
└── src/                   # reusable modules (env.py, agent.py, etc.)
...

```

12. FULL PIPELINE SUMMARY (for other AI)

1. **Data Ingestion**: load prices, Fama-French factors, alphas; load FNSPID news CSV, filter to 488 stocks, 2018-2020.
2. **BM25 ESG/SDG Tagging**: For each headline, compute BM25 against ESG term list and SDG term list; threshold to binary.
3. **FinBERT Sentiment**: Batch-predict positive/negative/neutral probabilities; compute tone.
4. **Daily Scores**: Multiply binary relevance × tone, average per stock per day → three score DataFrames (ESG, SDG, Overall).
5. **OLS Validation**: Regress next-day returns on lagged scores + market excess return; confirm positive significant coefficient.
6. **Stock Selection**: Rolling 6-month average sentiment → top 30 → select 10 with highest 3-factor alpha (from Step 0).
7. **RL Environment**: Build gym.Env with 240-dim state, Dirichlet action space, CVaR-based reward.
8. **PPO Training**: Train on 2018-2020 data, validate on 2021, test on 2022-2025.

9. ****Evaluation:**** Compute Sharpe, Sortino, max drawdown, turnover; compare to benchmarks; bootstrap significance.

10. ****Report:**** Write up methodology, results, discussion; include equations, tables, figures.

13. IMPORTANT TECHNICAL NOTES

- The FNSPID CSV has a known issue when loading via Hugging Face `datasets` (ArrowInvalid). We avoid this by direct HTTP download and pandas chunked reading.
- The reward CVaR must be computed from trailing portfolio returns, not from asset-level CVaR, to correctly capture portfolio tail risk.
- Transaction cost penalty uses L1 norm of weight change. We set $\tau = 0.001$ (1 bp per traded stock, matching paper).
- Policy output must be transformed to valid weights (softmax or Dirichlet). Dirichlet is preferred as it naturally enforces $\text{sum}=1$ and positivity.

Please pass this entire document to the other AI. It contains every decision, rationale, and step needed to generate the complete project code, notebook, and final report without losing any of our work.